

Zakhar Yagudin

Abu Dhabi, UAE — open to relocation / remote | zakhar.iagudin@gmail.com | +971 58 540 3613 | [LinkedIn](#) | [Google Scholar](#) | [Portfolio](#) | EN (Advanced) · RU (Native)

SUMMARY

Lead Robotics / Perception Engineer with **5+ years** building production 3D detection, sensor-fusion, and control stacks for autonomous trucks and racing platforms. Shipped **one of the first L4 autonomous freight deployments in Europe** (Ozon) and led the team that took **2nd place at A2RL Silver League 2025**, the world's premier autonomous-racing championship. **7 publications** at IEEE including ICRA, IROS, IV and arXiv (3 first-author, 75+ citations) on VLM-based driving, multi-sensor 3D detection, and end-to-end autonomy. Currently a PhD candidate at Skoltech working on diffusion-based forecasting and deep RL for high-speed racing, while leading a 10-engineer perception-planning-control team.

EXPERIENCE

Lead Computer Vision Engineer — Khalifa University, Team FlyEagle Jan 2025 – Present (*full-time*)
Autonomous racing — A2RL championship; perception, planning, control Abu Dhabi, UAE

- Led a **10-engineer team** building the full autonomy stack for an A2RL race car at **250+ km/h**; owned architecture, roadmap, sprint decomposition, code review, and CI/CD.
- Secured **2nd place at A2RL Silver League 2025** against **12 international university teams** by shipping a **30 FPS camera+LiDAR 3D-detection pipeline** with **<40 ms end-to-end latency**.
- Authored **3 publications** on real-time 3D detection and cross-domain racing perception; established lab's evaluation harness and reproducibility standards.

Computer Vision Researcher — 168ROBOTICS Jun 2024 – Present (*part-time*)
Multi-sensor 3D perception for autonomous platforms Moscow, Russia

- Designed and shipped a **LiDAR 3D-detection baseline from scratch**, reaching **NDS 80** on a **15k-frame internal dataset**; defined the data pipeline, model architecture, and evaluation protocol.
- Reduced inference latency **1.5×** via TensorRT INT8 quantization and CUDA-kernel tuning, enabling **90 FPS deployment on Jetson Orin**.

Software Engineer, Autonomous Driving — Ozon Tech Jan 2021 – May 2024 (*full-time*)
L4 autonomous freight trucks (MAN TGM, Volvo FH) Innopolis, Russia

- Owned the **lateral and longitudinal control module** (C++17) for MAN TGM and Volvo FH autonomous trucks; ran on STM32-based ECU at **~100 Hz** control loop.
- Achieved **~30 cm lateral tracking error at 80 km/h** on M-7 highway public-road trials; tuned MPC + CLF/CBF safety layer for high-mass dynamics.
- Shipped controller to revenue trucks logging **100+ km of autonomous miles on the M-7 corridor** — contributing to **one of the first L4 freight deployments in Europe**.

Founder & CEO — StraightenUp Sep 2024 – Present (*part-time*)
AI-powered wearable for posture monitoring — Skolkovo resident Moscow / Skolkovo Co-founded HealthTech wearable; defined product, R&D, and GTM. Secured Skolkovo residency; semifinalist at Sevan Startup Summit 2025.

CEO — Chain of Blocks 2024 – 2025 (*acquired*)
Mobile App based on Smart contracts in TON Russia Founded Blockchain company **acquired in 2025**.

SELECTED PUBLICATIONS

Full list (7 papers, 3 first-author, 75+ citations) on [Google Scholar](#).

- **Yagudin, Z.**, Tananakin, M., Guo, Z., Madjid, N.A., Mebrahtu, M., Dias, J., Khonji, M., Tsetserukou, D. *Perspective-Fusion: A Lightweight Real-time 3D Detector for Autonomous Driving*. **IROS 2025**.
- **Yagudin, Z.**, Guo, Z., Lykov, A., Konenkov, M., Tsetserukou, D. *VLM-Auto: VLM-based Autonomous Driving Assistant with Human-like Behavior and Understanding for Complex Road Scenes*. arXiv:2405.05885, 2024 (**57 citations**).
- **Yagudin, Z.**, Mebrahtu, M., Jin, R., Huang, J., Yue, Y., Tsetserukou, D., Dias, J., et al. *EagleVision: A Multi-Task Benchmark for Cross-Domain Perception in High-Speed Autonomous Racing*. arXiv:2604.11400, 2026.
- Guo, Z., Lin, X., **Yagudin, Z.**, Lykov, A., Wang, Y., Li, Y., Tsetserukou, D. *METDrive: Multimodal End-to-End Autonomous Driving with Temporal Guidance*. **ICRA 2025**.

EDUCATION

Skoltech — PhD Candidate, Robotics 2025 – Present · Moscow, Russia
Topic: Uncertainty-Aware Trajectory Planning in Autonomous Racing via Diffusion-Based Opponent Forecasting and Deep Reinforcement Learning.

Skoltech — MSc, Robotics 2023 – 2025 · Moscow, Russia
GPA 5.0/5.0. Focus: Autonomous Driving, Multi-modal Perception, Deep Learning. Thesis: 3D Object Detection using Vision Transformers and Sensor Fusion.

Innopolis University — BSc, Computer Science (Robotics) 2019 – 2023 · Innopolis, Russia
GPA 4.6/5.0. Thesis: Controller design for autonomous vehicles (MPC, CLF/CBF).

SKILLS

Languages: C/C++ (C++17), Python, CUDA.

DL & Deployment: PyTorch, TensorRT, ONNX, Hugging Face; INT8 quantization, real-time inference on Jetson Orin / STM32.

Perception: 3D Object Detection, Camera+LiDAR Sensor Fusion, BEV, Vision Transformers, VLM / VLA, multi-task perception.

Robotics & Control: ROS, MPC, CLF/CBF, behavior cloning, PPO/SAC, sim2real; Isaac Lab, MuJoCo, CARLA.

Leadership: Technical lead, roadmap & architecture ownership, hiring, code review, cross-functional alignment, founder experience.